

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COMPUTER ARCHITECTURE (CSA12)

Submitted By

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Q1 – Smart City Surveillance System

Introduction

Computer architecture studies how the CPU, memory, buses, and input/output devices work together to execute instructions efficiently. The given application is analyzed using concepts such as the fetch–decode–execute cycle, addressing modes, segmentation, memory hierarchy, and processor performance.

Discussion

The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications. The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications.

Process Flow

Camera → I/O Controller → System Bus → Main Memory → CPU → Threat Detection → Display

Code

```
LXI H, 2500H
MVI B, 05H
LOOP: MOV A, M
INX H
DCR B
JNZ LOOP
HLT
```

Conclusion

The architecture and assembly instructions demonstrate how efficient instruction execution and proper memory organization improve overall system performance.

Q2 – Stock Trading System

Introduction

Computer architecture studies how the CPU, memory, buses, and input/output devices work together to execute instructions efficiently. The given application is analyzed using concepts such as the fetch–decode–execute cycle, addressing modes, segmentation, memory hierarchy, and processor performance.

Discussion

The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications. The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications.

Process Flow

Fetch → Decode → Execute → Memory Access → Write Back

Code

```
LXI H, 3000H
MOV A, M
INX H
ADD M
STA 3050H
HLT
```

Conclusion

The architecture and assembly instructions demonstrate how efficient instruction execution and proper memory organization improve overall system performance.

Q3 – 8085 Temperature Controller

Introduction

Computer architecture studies how the CPU, memory, buses, and input/output devices work together to execute instructions efficiently. The given application is analyzed using concepts such as the fetch–decode–execute cycle, addressing modes, segmentation, memory hierarchy, and processor performance.

Discussion

The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications. The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications.

Process Flow

Sensor → IN → Compare → OUT → Actuator

Code

```
IN 00H
CPI 32H
JC LOW
MVI A, 01H
OUT 01H
HLT
LOW: MVI A, 00H
OUT 01H
HLT
```

Conclusion

The architecture and assembly instructions demonstrate how efficient instruction execution and proper memory organization improve overall system performance.

Q4 – 8086 Segmentation

Introduction

Computer architecture studies how the CPU, memory, buses, and input/output devices work together to execute instructions efficiently. The given application is analyzed using concepts such as the fetch–decode–execute cycle, addressing modes, segmentation, memory hierarchy, and processor performance.

Discussion

The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications. The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications.

Process Flow

CS:IP → Fetch → Decode → DS → Execute → SS

Code

```
MOV AX, DATA
MOV DS, AX
MOV AX, NUM
PUSH AX
POP BX
MOV AH, 4CH
INT 21H
```

Conclusion

The architecture and assembly instructions demonstrate how efficient instruction execution and proper memory organization improve overall system performance.

Q5 – Number Systems and Benchmarking

Introduction

Computer architecture studies how the CPU, memory, buses, and input/output devices work together to execute instructions efficiently. The given application is analyzed using concepts such as the fetch–decode–execute cycle, addressing modes, segmentation, memory hierarchy, and processor performance.

Discussion

The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications. The processor continuously communicates with memory and peripheral devices through the system bus. Efficient bus organization, cache memory, DMA, instruction pipelining, and proper addressing modes reduce execution time and improve throughput. These concepts are essential for designing reliable computer systems and embedded applications.

Process Flow

Hexadecimal → Binary → CPU Decode → Execute → Output

Code

```
MOV AL, 3AH
MOV BL, AL
AND BL, 0FH
MOV CL, 04H
SHR AL, CL
HLT
```

Conclusion

The architecture and assembly instructions demonstrate how efficient instruction execution and proper memory organization improve overall system performance.